1.31 show that $H[X,Y] \le H[X] + H[Y]$, with equality

$H[X,Y] = H[X] + H[Y] \iff X, Y$ are independent

1) $\Rightarrow$

$H[X,Y] = H[Y|X] + H[X] = H[Y] + H[X] - I[X,Y] = H[X] + H[Y]$

if $I[X,Y] = 0 \Rightarrow$

$\Rightarrow \quad 0 = I[X,Y]$

$KL(p(X,Y) \| p(X)p(Y)) = 0$

$\iff p(X,Y) = p(X)p(Y)$

$\Rightarrow$ X, Y are independent.

2) $\Leftarrow$

X, Y are independent $\Rightarrow \; I[X,Y] = H[Y] - H[Y|X]$

$= H[Y] - \underline{H[Y|X] - H[X]} + H[X]$

$= H[Y] + H[X] - H[X,Y] = H[Y] + H[X] + \iint p(x,y) \ln p(x,y)\, dx\, dy$

$= H[Y] + H[X] + \iint p(x,y) \ln p(x)\, dx\, dy + \iint p(x,y) \ln p(y)\, dx\, dy$

$= H[Y] + H[X] - (-\int p(x) \ln p(x)\, dx) - (-\int p(y) \ln p(y)\, dy) = 0.$

thus, $H[X,Y] = H[Y] + H[X] \iff$ X, Y are independent.

3) $H[X,Y] \le H[X] + H[Y]$

$\because I[X,Y] = KL(p(x,y) \| p(x)p(y)) \ge 0$ by Jensen inequality

$I[X,Y] = H[Y] + H[X] - H[X,Y] \ge 0.$

$\therefore H[X,Y] \le H[Y] + H[X]$, X, Y are independent

iff $H[X,Y] = H[Y] + H[X]$ by 1, 2, 3